

WHAT LIES BEYOND THE CUT

How much turbulence can a coarse grid know about?

Approaching the optimal closure: equivariance, inductive bias, and Reynolds-number generalization in data-driven LES

Syver Døving Agdestein · Benjamin Sanderse

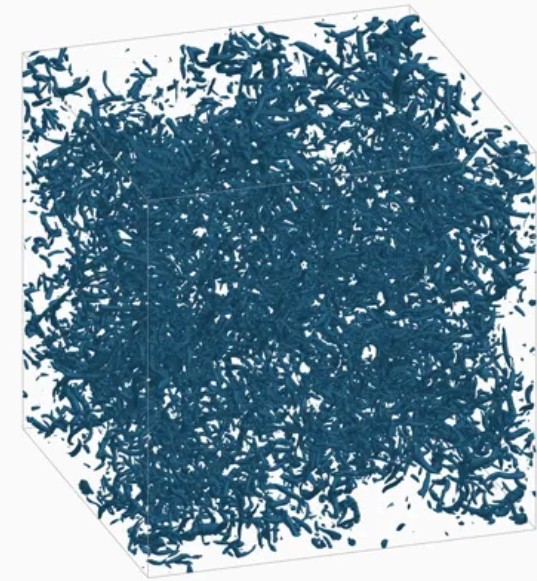
Centrum Wiskunde & Informatica, Amsterdam · Eindhoven University of Technology

WCCM–ECCOMAS 2026 · Munich · 22 July

MS184A · Theory-guided Design of Deep Learning-based Surrogates

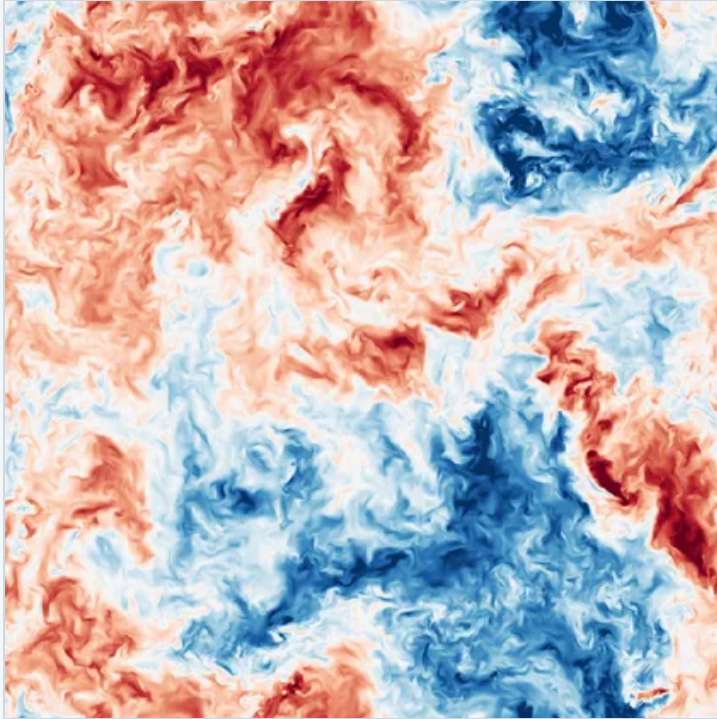
Funded by NWO Talent Programme Vidi (VI.Vidi.193.105)

compute on SURF / Snellius (EINF-15798)

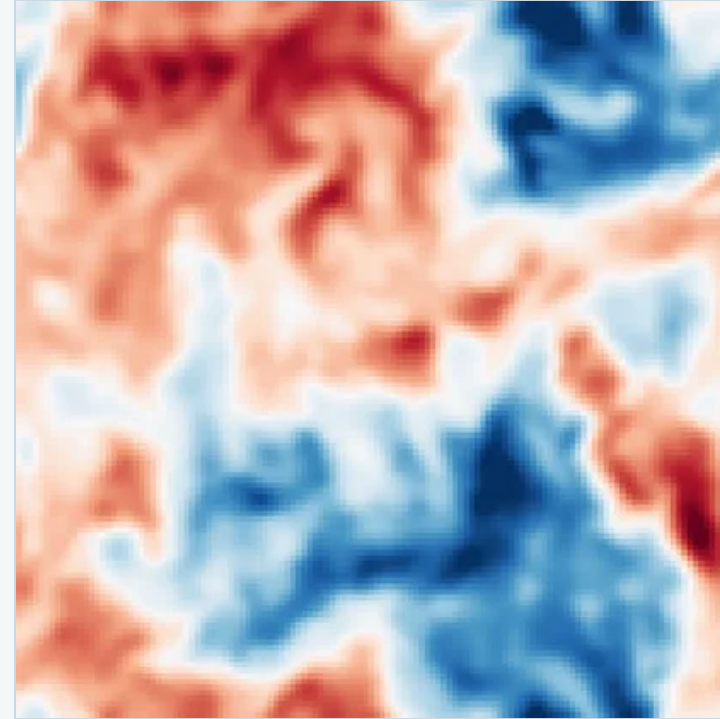


decaying Taylor–Green vortex at $Re = 3000$

The same flow, on two grids



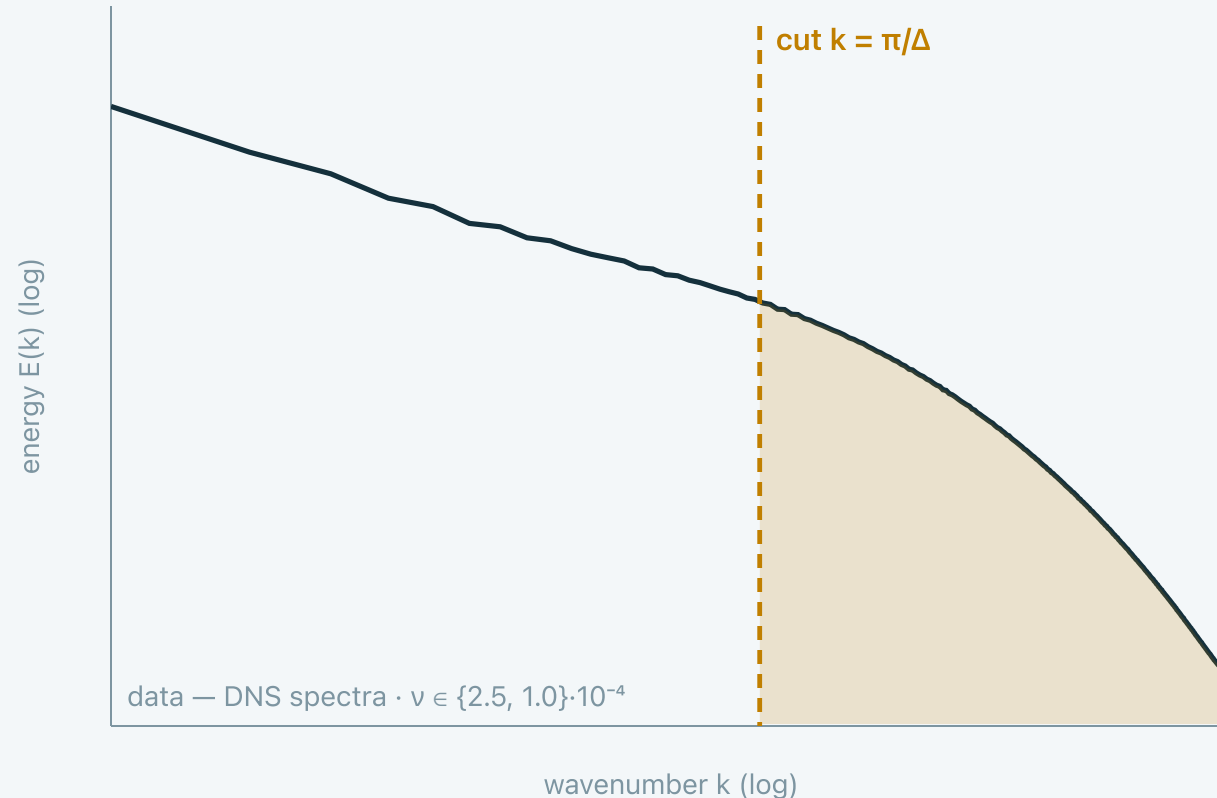
DNS · VELOCITY u
810³ — resolves everything



LES · FILTERED VELOCITY $\bar{u}$
128³ — 250× fewer points, misses the small scales

the missing scales still act on the resolved ones — that action is the closure's job

Where the energy lives — and where the grid stops



the grid resolves everything left of the cut;
the shaded energy is invisible

coarsen the filter ($\Delta \uparrow$) — more energy is hidden

lower the viscosity ($\nu \downarrow$) — the tail grows longer

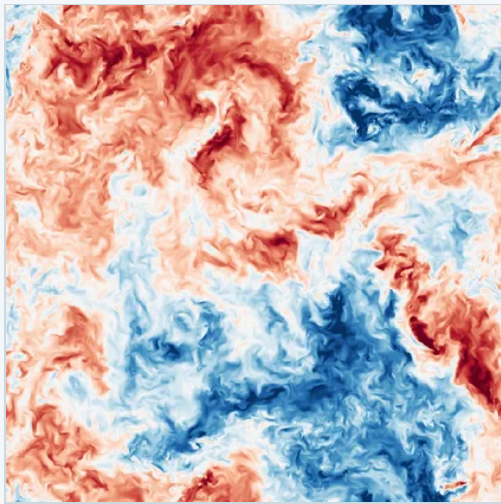
$$\text{Re}_\Delta = \frac{\Delta^2 \|\bar{\bar{A}}\|}{\nu}$$

one resolved-scale number for

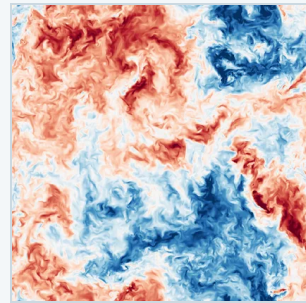
how much lies beyond the cut

$\bar{\bar{A}} = \nabla \bar{u}$ — the resolved velocity gradient ·
 $\|\bar{\bar{A}}\|$ — its r.m.s. over the box

Why this number? Scale the equations and see what survives



a turbulent solution



...rescaled – still a solution

incompressible Navier–Stokes

$$\nabla \cdot u = 0, \quad \partial_t u + \nabla \cdot (uu + pI - \nu \nabla u) = f$$

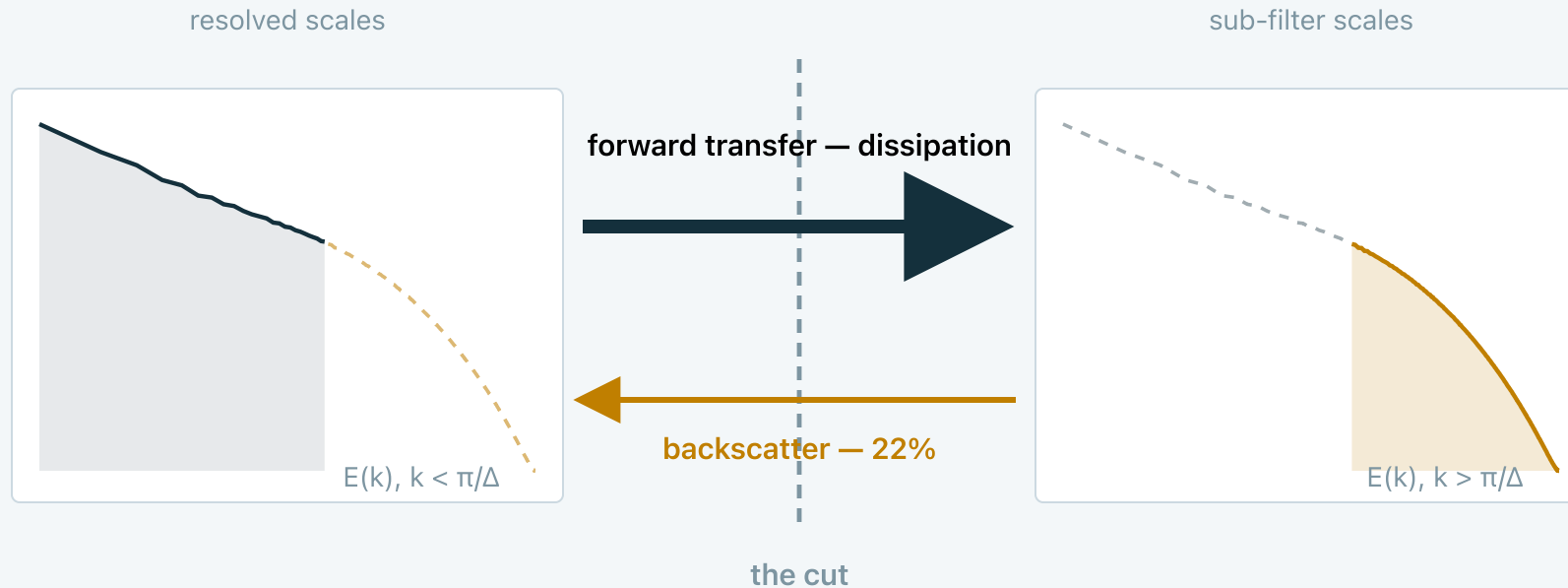
rescale space and time

$$x \rightarrow ax, \quad t \rightarrow bt, \quad u \rightarrow \frac{a}{b} u, \quad \nu \rightarrow \frac{a^2}{b} \nu$$

Still Navier–Stokes — but the rescaling multiplies ν by a^2/b . A closure blind to ν stays consistent only where $\mathbf{a}^2 = \mathbf{b}$, the scalings that leave ν fixed.

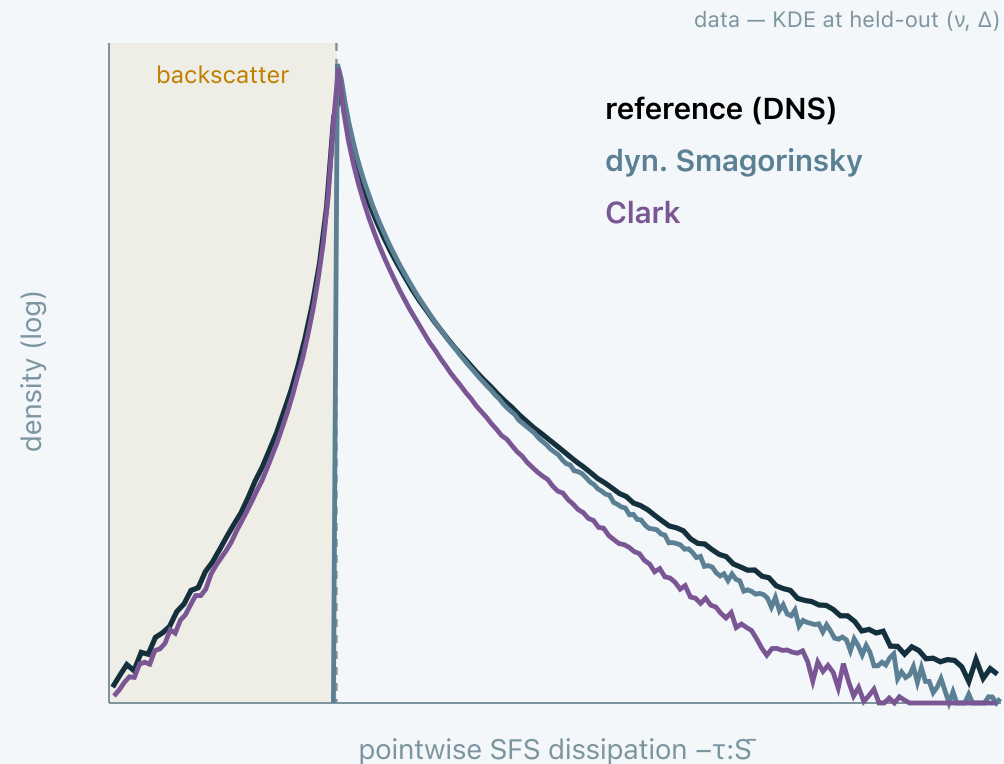
Along that family, $\bar{u}$ has a single dimensionless invariant: Re_Δ . Symmetry itself hands the closure its Reynolds input.

Energy crosses the cut in both directions



$\Pi = -\tau : \bar{S}$ a closure must get this net flux right — at every Reynolds number

Classical closures pick one side of this trade



FUNCTIONAL · SMAGORINSKY

$$\tau = -2 (C_s \Delta)^2 |\bar{S}| \bar{S}$$

dissipation $\approx$ right · structure wrong

backscatter 0.0004 – can't flow backward

STRUCTURAL · CLARK

$$\tau = \frac{\Delta^2}{12} \bar{A} \bar{A}^T$$

structure $\approx$ right · dissipation 0.6× too weak

backscatter 0.25 – close to the true 0.22

and notice: **v** appears in neither formula

no v

Smagorinsky 1963 · Germano et al. 1991 · Clark et al. 1979

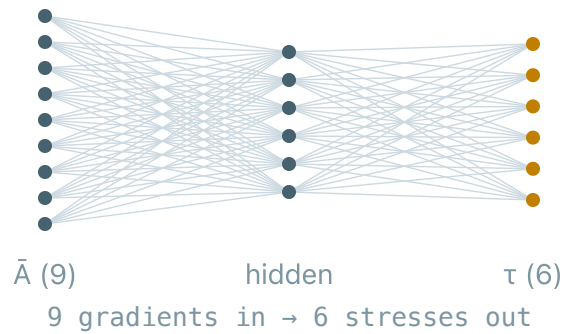
Learn the closure from DNS — building in the physics we know

$$m : \underbrace{\bar{A}(x)}_{\mathbb{R}^{3 \times 3}} \mapsto \underbrace{\tau(x)}_{\text{sym. deviatoric}}$$

one network, fit a priori (least squares on τ) · 3 viscosities × 3 filter widths · 5 seeds

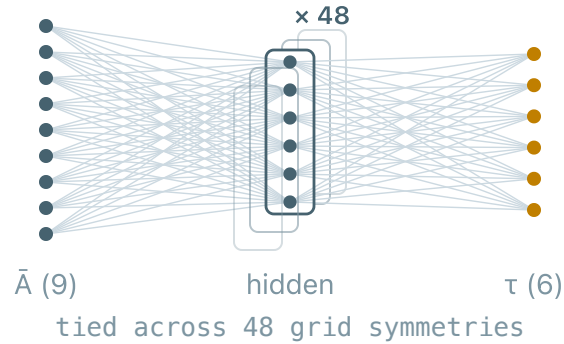
more inductive bias built in — less for the network to learn

MLP · NO SYMMETRY



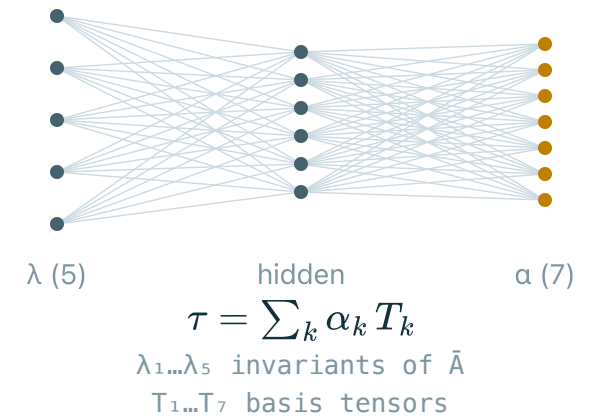
structure must be **learned** from data
Rosenblatt 1958

G-CNN · TIED WEIGHTS



equivariant by **weight sharing**
Cohen & Welling 2016

TBNN · TENSOR BASIS



equivariant through the **features**
Pope 1975 · Ling et al. 2016

However much we build in, all three stay **single-point closures** — each reads $\tau(x)$ from $\bar{A}(x)$ alone. Hold that thought.

The learned closures are just as blind

WITHOUT $\mathbf{v}$, THE SCALING SYMMETRY FORCES

$$\tau = \Delta^2 |\bar{A}|^2 m\left(\bar{A}/|\bar{A}|\right) \quad \text{no } \mathbf{v}$$

consistent — but **rigid three times over**:

τ can only grow as Δ^2

τ can only grow as $|\bar{A}|^2$

τ never feels $\mathbf{v}$ — calibration frozen at the training regime

let $\mathbf{v}$ back in — through the one dimensionless number the symmetry permits

SAME SYMMETRY, ONE EXTRA INPUT

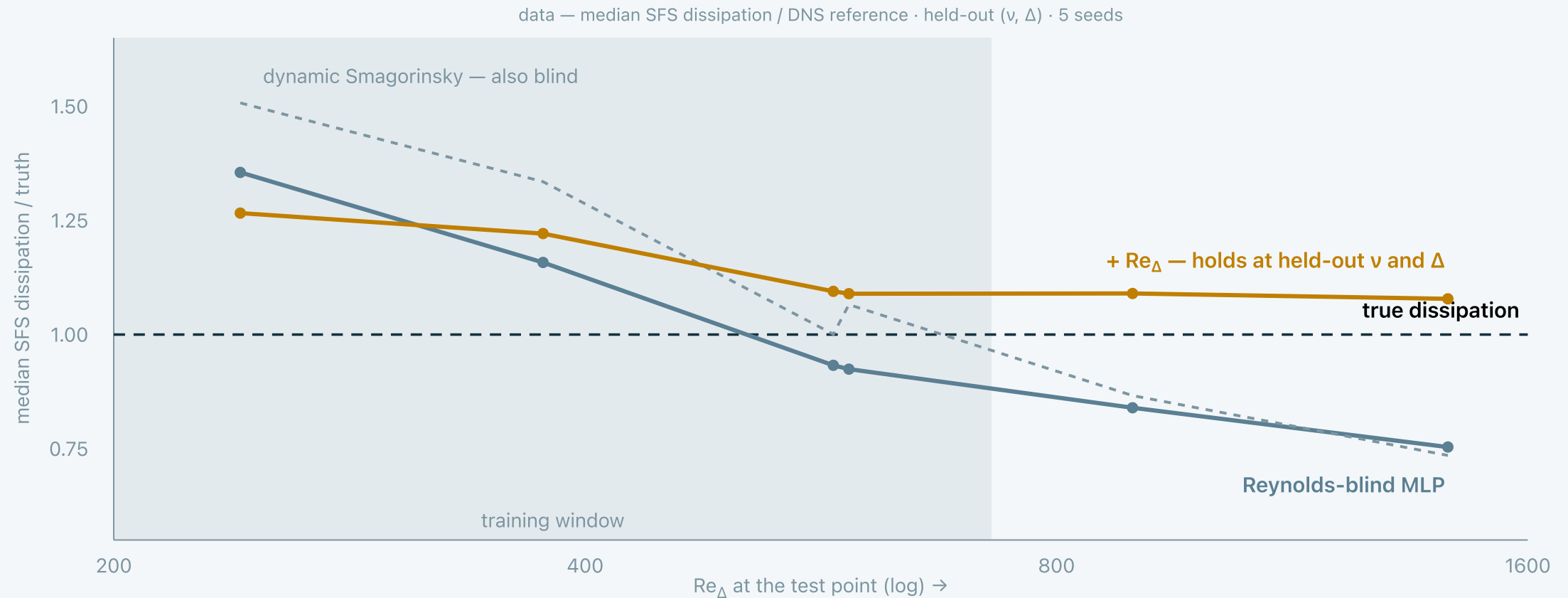
$$\tau = \Delta^2 |\bar{A}|^2 m\left(\bar{A}/|\bar{A}|, \text{Re}_\Delta\right) \quad \text{with } \mathbf{v}$$

now m bends **nonlinearly** with Δ , $|\bar{A}|$ and $\mathbf{v}$ —

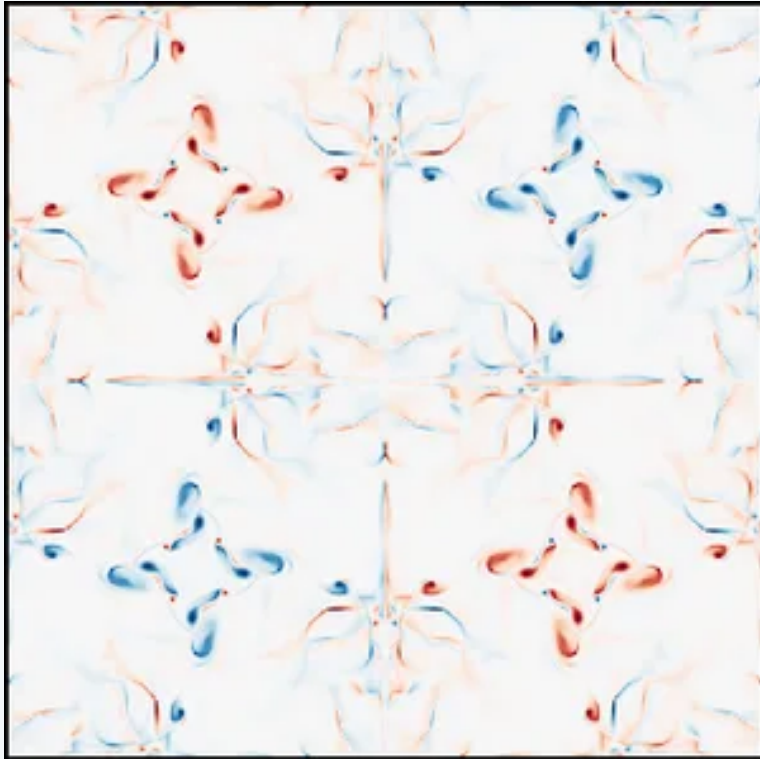
but only through $\text{Re}_\Delta = \Delta^2 \|\bar{A}\|/\mathbf{v}$,

the one combination the rescalings leave alone

Tested away from training: who keeps their calibration?



A flow it has never seen: Taylor–Green becoming turbulent

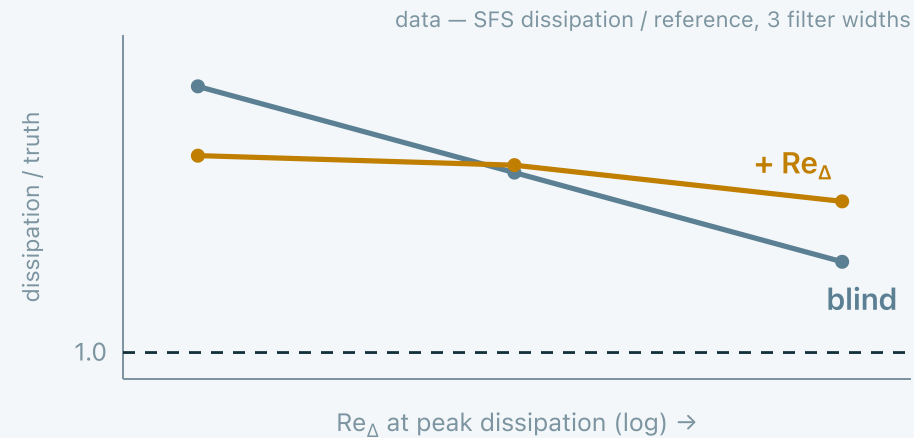


VORTICITY · $\mathbf{t} = \mathbf{o}$

decaying, transitional – out of distribution twice over

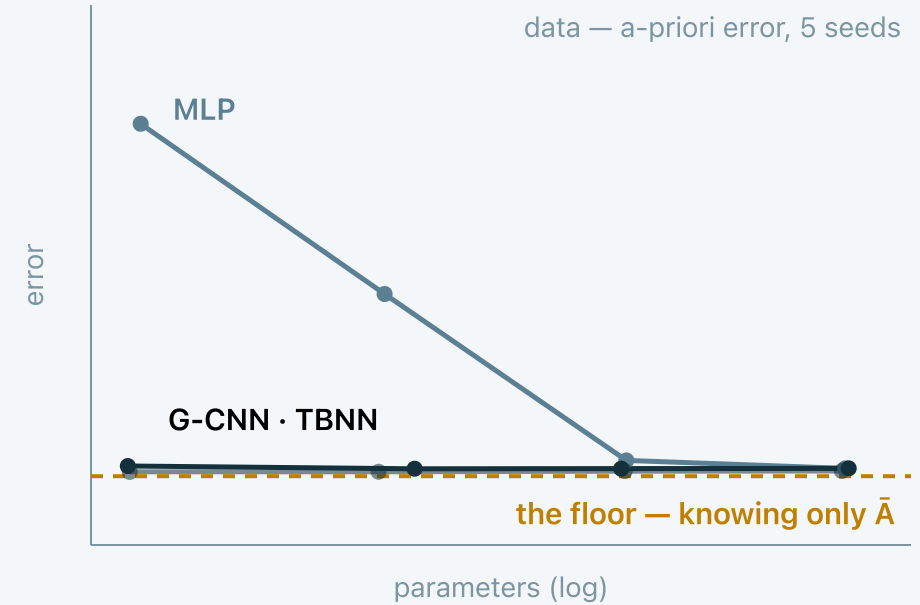
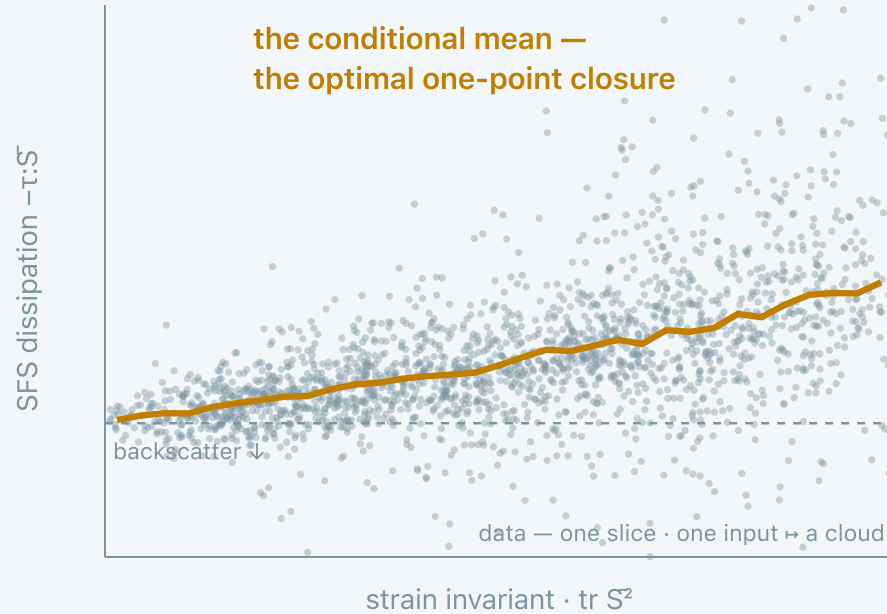
laminar → transition → decay

all closures stay stable — but over-dissipate



Re_Δ corrects the Reynolds-driven share of the excess,
and honestly leaves the share owed to different flow physics

Couldn't a better architecture have fixed this? We checked.



THE OPTIMAL CLOSURE

$$\min_m \mathbb{E} \|m(\bar{A}) - \tau\|^2 \implies m^*(\bar{A}) = \mathbb{E} [\tau \mid \bar{A}]$$

the best **any** one-point model can do

Langford & Moser 1999

ARCHITECTURE

how cheaply you reach the floor — 25 $\times$ cheaper, never below

INPUT

where the floor is — more input features can only lower it

$$\mathbb{E} \|m^*(\bar{A}, \text{Re}_\Delta) - \tau\|^2 \leq \underbrace{\mathbb{E} \|m^*(\bar{A}) - \tau\|^2}_{\text{the floor}}$$

TAKEAWAY

A closure that knows *what it's missing*
works where it has never been.

Agdestein & Sanderse · CWI, Amsterdam · TU Eindhoven

*"Approaching the optimal closure: equivariance, inductive bias,
and Reynolds-number generalization in data-driven LES"*

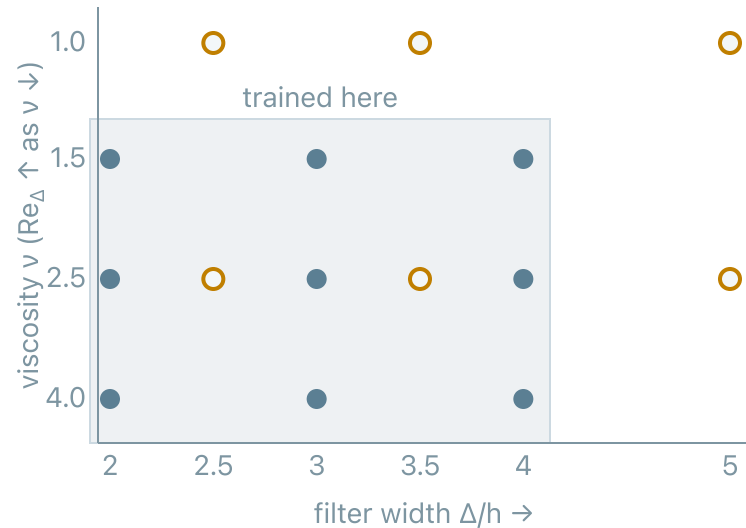
arxiv preprint: **2603.05325** · code on GitHub



Read the interactive story of this work
<https://agdestein.github.io/>

Setup: data and training

EXPERIMENT DESIGN



● trained (3×3) · ○ tested – held-out seed & ν , off-grid Δ

every test point sits off the training grid — in filter width *and* in Re_Δ
 train $Re_\Delta \approx 104\text{--}727$ · 8 snapshots per grid point $\rightarrow$ 72 gradient-stress fields

DATA & TRAINING

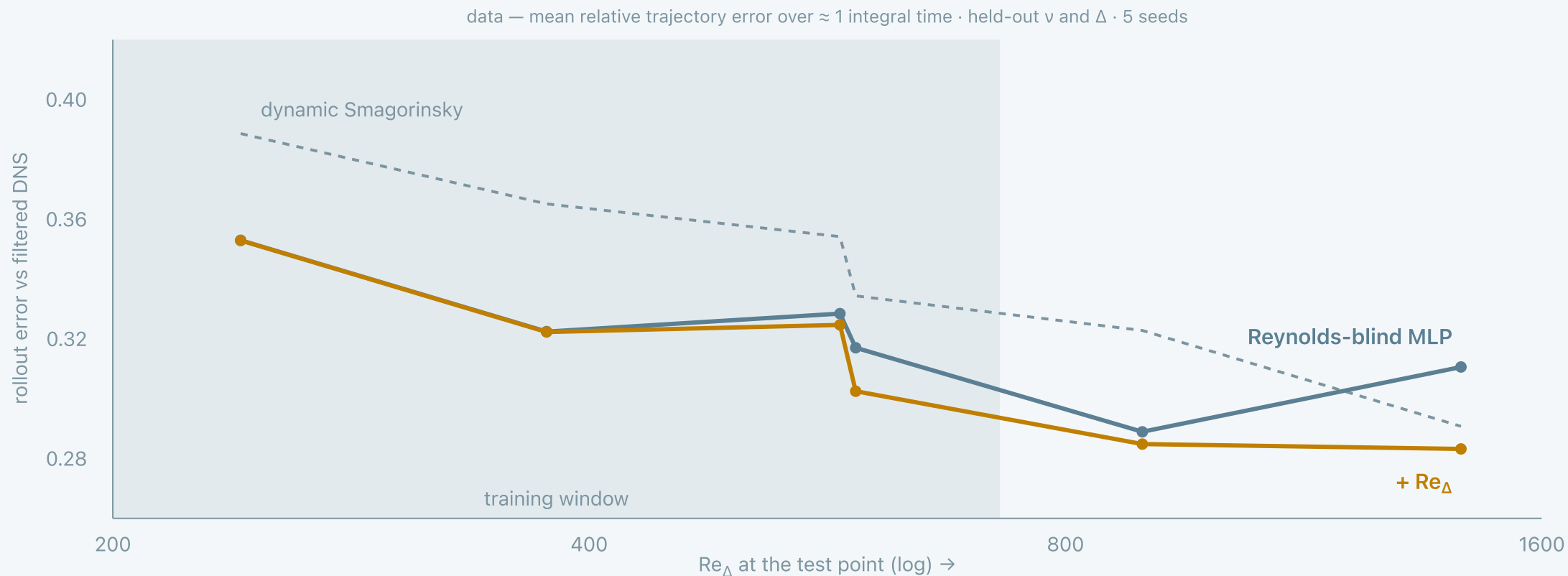
DNS — forced HIT, pseudo-spectral 810^3 · one H100
 2/3-rule dealiasing · $k_{\max}\eta \geq 1.1$ · double precision

Filter — modified Gaussian onto a 128^3 LES grid
 kernel commutes with the rescaling symmetry

Closures — MLP · G-CNN · TBNN, four sizes ($\approx 120\text{--}3000$ params)
 5 seeds each · a-priori least squares on the normalized stress · AdamW, 20 epochs

A posteriori — rollouts vs 40 filtered-DNS snapshots (≈ 1 integral time)
 error = mean relative trajectory error · baselines: no-model · dynamic Smagorinsky · Clark

The calibration gain survives an actual LES run



inside the training window the two are essentially tied; the **Re_Δ input** pulls ahead at the highest Re (−9% at the farthest test point)

Scoreboard at the in-distribution test point ($\Delta/h = 2.5$)

CLOSURE	A-PRIORI STRESS ERROR	A-POSTERIORI ROLLOUT ERROR
No-model	1.000	0.432
Dynamic Smagorinsky	0.964	0.389
Clark	0.466	0.364
MLP	0.449	0.353
G-CNN	0.449	0.355
TBNN	0.446	0.351
conditional-mean floor	0.439	—

a priori, the data-driven closures sit on the one-point floor; a posteriori the spread narrows but the ranking holds — the learned closures stay ahead of every classical baseline

learned closures: largest tier (≈ 3000 parameters), median over 5 seeds · all metrics at held-out seed $v = 2.5 \cdot 10^{-4}$, $\Delta/h = 2.5$